

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.4.2 (prototype)
Log entry date	9 October 2026
Status	Released: pushed to the GitHub main branch on 9 October 2026, in the commit that adds this log; Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch (commits titled Release v0.4.2)
Previous version	0.4.1, 8 October 2026, commit fe06178 (see Spaghetti_Engine_Dev_Log_v0.4.1.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.4.1 (8 October 2026) to version 0.4.2 of The Spaghetti Engine. Version 0.4.2 is a small release: it makes the interface easier to read, and it adds one column to the results of the Transition Point Analysis tab. No analysis is computed differently.

The log documents what changed and why, how it was checked, and the deployment status. It also records the deployment of 0.4.1, which the 0.4.1 log published on the site still described as not yet deployed.

2. Summary of Version 0.4.2

Area	Change
Readability	Text on light backgrounds is black or very dark grey instead of light grey, and the small text sizes are larger: 10 to 11.5 pixels, 11 to 12.5, 12 to 13, and 14 to 15. Chart axis labels are darker and 12 pixels. Light text is kept on the dark header bar and the dark box at the top of the About page.
Change at TP	A new column in the Transition Point results table, its CSV export, and the Analysis Workbook: the change in the outcome from intensity 0 to the end of the steepest step, which begins at the transition point: $Y(TP + \Delta x) - Y(0)$, next to Max Change, the change at full intensity.
Influence Metrics	The cells of the module matrix with a strong colour show white text, so their numbers stay readable.
Record of 0.4.1	Version 0.4.1 was pushed as commit fe06178 on 8 October 2026 and deployed the same day.

3. The Changes in Detail

3.1 Readability

- **Text colour.** Every grey text class on a light background was replaced: the darker greys by near-black (neutral-900), the mid grey used for hints by dark grey (neutral-700), and the lightest greys by neutral-600 and neutral-500. Hover states turn black. Text on the dark header bar and the About page's dark box keeps its light colour, slightly brightened.
- **Text size.** The fixed small sizes step up (10 to 11.5 pixels, 11 to 12.5), and the two text sizes the interface uses most are larger for the whole site (12 to 13 pixels, 14 to 15), set once in the theme of src/index.css. On-screen chart ticks are 12 pixels, and the chart axis labels use near-black instead of the charting library's light grey.
- **Unchanged.** The default font sizes of exported figures (the figure panel), which are chosen for print at journal sizes, and the brand colours.

3.2 Change at TP

- **What it shows.** For each outcome, how far it has moved once the steepest step, which begins at the transition point, is complete: its value one step past the TP minus its value at intensity 0, measured the same way as Max Change, which runs from intensity 0 to full intensity.
- **Reading it.** Compared with Max Change, it shows how much of the full effect is reached by implementing the intervention just past its transition point. Because the steepest step is included, a TP of 0 still shows the change that this first step brings.
- **Where.** The results table (sortable, between TP and Max Change), the CSV export (column Change at TP), and the Transition Point Analysis sheet of the Analysis Workbook (column Change at Transition Point).

4. Implementation Choices

Topic	Choice in the tool	Reason
Text hierarchy	Main text black, hints dark grey.	A slight difference keeps headings, values, and hints distinguishable; every text stays well above the contrast standard.
How sizes change	The theme's two most used sizes are enlarged once, and fixed sizes are stepped up.	One change applies throughout the interface, and no size ends below 11.5 pixels.
Change at TP	$Y(TP + \Delta x) - Y(0)$: up to the end of the steepest step.	The steepest step is the change the transition point marks, so it belongs in the value. The first push of 0.4.2 used $Y(TP) - Y(0)$, which showed 0 for every TP of 0; it was replaced the same day, within version 0.4.2.
Figure defaults	Not changed.	Figure sizes and fonts are set for print; the figure panel already lets them be adjusted.

5. Effect on Existing Results

- No analysis is computed differently; results from 0.4.1 still hold.
- The Transition Point CSV export and the workbook sheet have one more column, placed after the transition point; scripts that read these files by column position need adjusting.
- Version raised from 0.4.1 to 0.4.2 in package.json and package-lock.json.

6. Defects Found and Fixed

Problem	Cause	Fix
In the module matrix of Influence Metrics, numbers in strongly coloured cells became hard to read once the text was black.	The cells inherit the table's text colour, whatever their background.	Each cell picks white or black text from the luminance of its colour blended at the cell's opacity.
The 0.4.1 log on the site still said that release was not yet deployed.	The log was written before the push, and the corrected log was not pushed.	This log records the 0.4.1 deployment (Section 2).

7. Verification

- **Tests:** tests/test_transition_direction.cjs now has 33 checks, all passing; eight new ones confirm that Change at TP equals $Y(TP + \Delta x) - Y(0)$ and equals a run with the lever held one step past the TP minus a run with it held at 0, for increases and decreases, on the water-energy-food and peat meadow examples.
- **Contrast:** in the browser, every visible text element was measured against its background on the About page and the Model editor, Network, Influence Metrics, Scenarios & Simulation, Monte Carlo, and Transition Point Analysis tabs; all reach at least 4.5:1 (the WCAG AA level for body text).
- **Layout:** no text is smaller than 11.5 pixels except the small tagline under the logo in the About page's dark box; the page does not scroll sideways at 1024 pixels wide, where the header controls wrap onto two lines.
- **Build:** the production build completes without errors.

8. Deployment Status

- **Deployed** with the pushes of this release to main on 9 October 2026; the second one changed the definition of Change at TP (Section 4). Netlify rebuilds the site automatically.
- **Files changed:** src/SpaghettiEngine.jsx, src/AboutTab.jsx, src/index.css, tests/test_transition_direction.cjs, package.json and package-lock.json, README.md, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.4.2 versions replace those of 0.4.1.
- **Build note:** npm run build fails in the local OneDrive folder because npm's command shim cannot handle the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 and 0.4.1 logs still apply.
- Change at TP depends on the sweep resolution, because the steepest step is one grid step wide.

- Below about 1,100 pixels wide, the header controls take two lines.
- The regression tests of 0.4.0 are still not in the repository; only tests/test_transition_direction.cjs is.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.4.2)	How to use every feature. Section 1.2 lists the changes of 0.4.2; Section 12.5 describes the Change at TP column.
The Spaghetti Engine - Methodological Annex (version 0.4.2)	The methods. Section 12.3 defines the change at the transition point.
Spaghetti_Engine_Dev_Log_v0.4.1	The 0.4.1 release: a decrease runs from 0 to -1 on the Transition Point and Transition Pathways tabs.
Spaghetti_Engine_Dev_Log_v0.4.0	The 0.4.0 release: the Monte Carlo tab, tab groups, the new branding, and the About & documentation tab.
This Development Log	What changed in 0.4.2 and the technical work behind it.

11. Next Steps

- Rebuild the full engine regression set inside tests/.
- Continue with the development roadmap: version 0.5.0 (guided first use) is next.